Monopolistic Competition: Choice and Entry

Differentiated products → price and output → entry and exit → welfare and advertising

1. Between perfect competition and monopoly

Perfect competition and monopoly are useful benchmarks. Many actual markets lie between them. There may be numerous sellers, yet each offers a product that is somewhat different from the others. A firm then has some control over its own price, even though customers can switch to rivals.

> **DEFINITION Monopolistic competition**
>
> **Monopolistic competition** is a market structure with **many sellers**, **differentiated products**, and **free entry and exit** in the long run. Each firm faces a downward-sloping demand curve for its own product: raising its price loses some customers, but not necessarily all of them.

Differentiation can arise from product features, reliable service, reputation, or location. Consider independent firms supplying monthly payroll-processing services to small businesses. Their services overlap, but some customers prefer particular reporting features or integration arrangements. No single supplier controls the whole market; nevertheless, each has customers who would not immediately switch for a small price difference.

Market structure	Sellers and products	Price and entry
Perfect competition	Many; effectively identical goods	Price taking; entry and exit free
Monopolistic competition	Many; differentiated varieties	Some pricing discretion; entry and exit free
Oligopoly	Few important rivals; goods may differ	Decisions depend on rivals; entry may be difficult
Monopoly	One seller without close substitutes	Market power maintained by entry barriers

An **oligopoly** is a market dominated by a few important sellers whose decisions affect one another. A **four-firm concentration ratio** is the combined share of market sales made by the four largest firms; for example, four shares of 20%, 15%, 10%, and 5% sum to 50%. Chapter 18 examines oligopoly in detail.

> **KEY RESULT Two features belong together**
>
> Monopolistically competitive firms have **market power over their own variety** because products differ. But **free entry and exit** make persistent positive economic profit difficult. In the short run, behavior resembles monopoly; in the long run, entry and exit bring economic profit toward zero.

So what? A differentiated firm is neither a pure price taker nor a protected monopolist. A firm can charge a markup over marginal cost and still fail to earn positive economic profit after entry has occurred. The distinction matters for the graphs that follow.

2. Short-run equilibrium

A differentiated seller faces a downward-sloping demand curve for its own service. Selling an extra contract generally requires a lower price, including on contracts that would otherwise have been sold at the higher price. As in monopoly, marginal revenue lies below the price read from demand.

> **DEFINITION Revenue, output, and profit**
>
> Total revenue is $TR = P \times Q$; average revenue is $AR = TR/Q = P$. Marginal revenue (MR) is the extra revenue from one more unit. Marginal cost (MC) is the extra cost of that unit. A firm increases output while additional revenue exceeds additional cost. On a smooth graph, its best positive output satisfies $\boxed{MR = MC}$; **price is then read from demand**, not from MR.

For the diagrams, consider a hypothetical payroll-service firm. The firm pays a fixed monthly cost of \$64 and has total cost $TC = 64 + 4Q + 0.25Q^2$. Its average total cost is $ATC = 4 + 64/Q + 0.25Q$, and its marginal-cost schedule is represented by $MC = 4 + 0.5Q$. These are *model equations* for drawing the curves, not formulas students must derive.

The firm initially faces demand $P = 24 - 0.75Q$. The corresponding smooth marginal-revenue curve is $MR = 24 - 1.5Q$. The vertical scale is dollars per monthly contract and Q is the number of contracts. These functions are held fixed whenever the same demand situation appears in another figure.

Worked example: Why marginal revenue is below price

For the same demand schedule, the price must fall to sell more contracts. Multiplying each price by quantity gives total revenue. Comparing consecutive rows gives the *additional* revenue from the last contract:

Contracts Q	Price P	Revenue $TR = P \times Q$	Extra revenue MR
8	\$18.00	\$144.00	–
9	\$17.25	\$155.25	\$11.25
10	\$16.50	\$165.00	\$9.75
11	\$15.75	\$173.25	\$8.25

From 9 to 10 contracts, revenue grows by only 165.00 − 155.25 = \$9.75 even though the 10th contract sells for \$16.50. The lower price also reduces revenue on the previous 9 contracts. That *price effect* explains why marginal revenue is below price. The graph uses smooth curves; table marginal revenues measure exact whole-contract changes and therefore need not equal the smooth curve's value at an endpoint.

> **KEY RESULT The short-run decision**
>
> First choose output where $MR = MC$, on the upward-sloping part of MC. Then go **up to demand** to read price. At that quantity, if $P > ATC$, the firm earns economic profit; if $P < ATC$, it makes a loss. A loss-making firm may nevertheless continue producing in the short run if revenue covers avoidable operating costs.

Worked example: A service supplier earns a short-run profit

At $Q = 10$, $MR = 24 - 1.5(10) = \$9$ and $MC = 4 + 0.5(10) = \$9$. Demand gives the selling price $P = 24 - 0.75(10) = \$16.50$. Average total cost is $4 + 64/10 + 0.25(10) = \$12.90$. Thus:

$$\text{Profit} = (16.50 - 12.90) \times 10 = \boxed{\$36}.$$

The price exceeds average cost by \$3.60 for each of 10 contracts. This is a profit rectangle, not the area between demand and marginal cost.

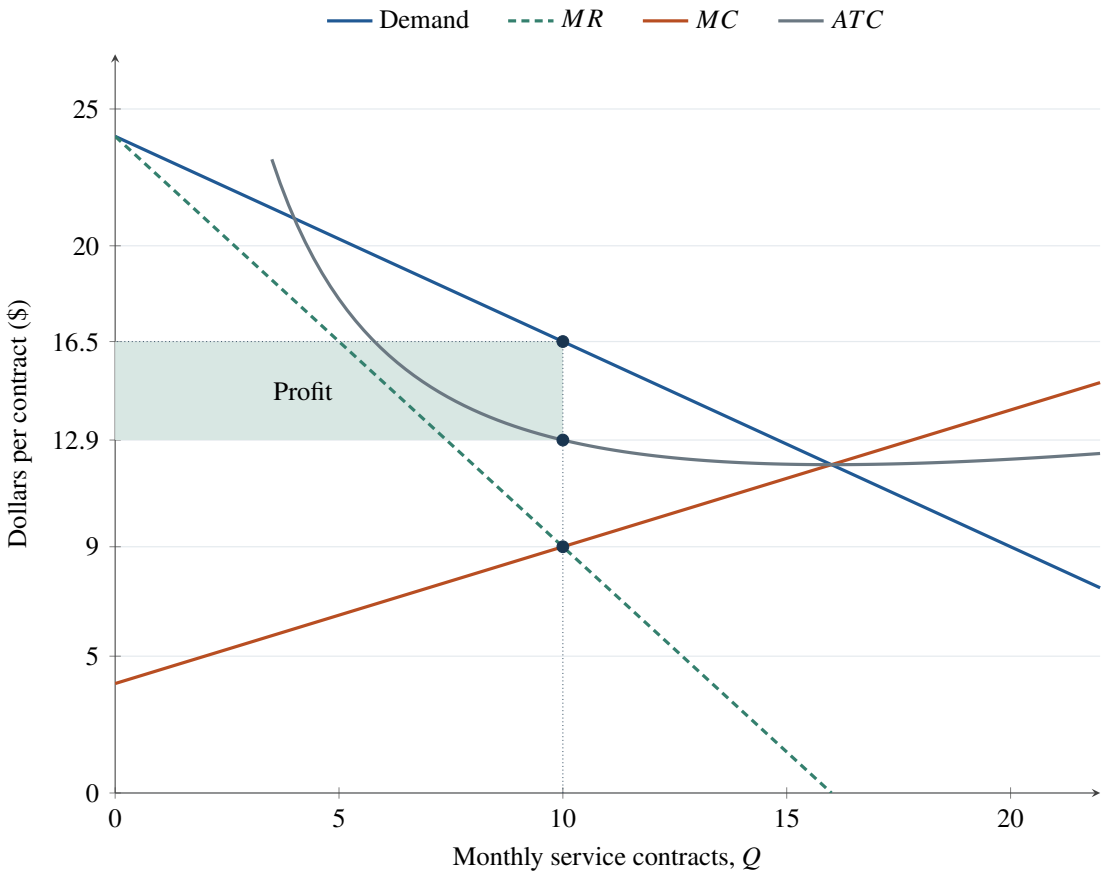


Figure 1. Short-run economic profit. The firm produces 10 contracts where $MR = MC = \$9$, charges \$16.50 from demand, and has $ATC = \$12.90$. The shaded rectangle is \$36.

Read the graph. Begin where the dashed green MR line crosses the orange MC line, at $Q = 10$. Move vertically to the blue demand curve to find \$16.50. Then locate ATC at *the same output*: \$12.90. The gap is positive profit per unit; multiplying by quantity gives total profit.

Worked example: The same cost structure can produce a short-run loss

Suppose rival suppliers become more attractive and this firm's demand instead is $P = 16 - 0.75Q$, with $MR = 16 - 1.5Q$. The profit-maximizing positive output is now $Q = 6$ because $MR = MC = \$7$. The firm charges $P = \$11.50$. Its average cost is $ATC = 4 + 64/6 + 0.25(6) = \$16\frac{1}{6}$.

Revenue is $6(11.50) = \$69$. Total cost is $64 + 4(6) + 0.25(36) = \$97$. Thus its **loss is \$28**. If it stops temporarily, it still owes \$64 in unavoidable fixed cost. Its average variable cost at 6 contracts is $AVC = 4 + 0.25(6) = \$5.50$, below price; operating reduces its loss from \$64 to \$28.

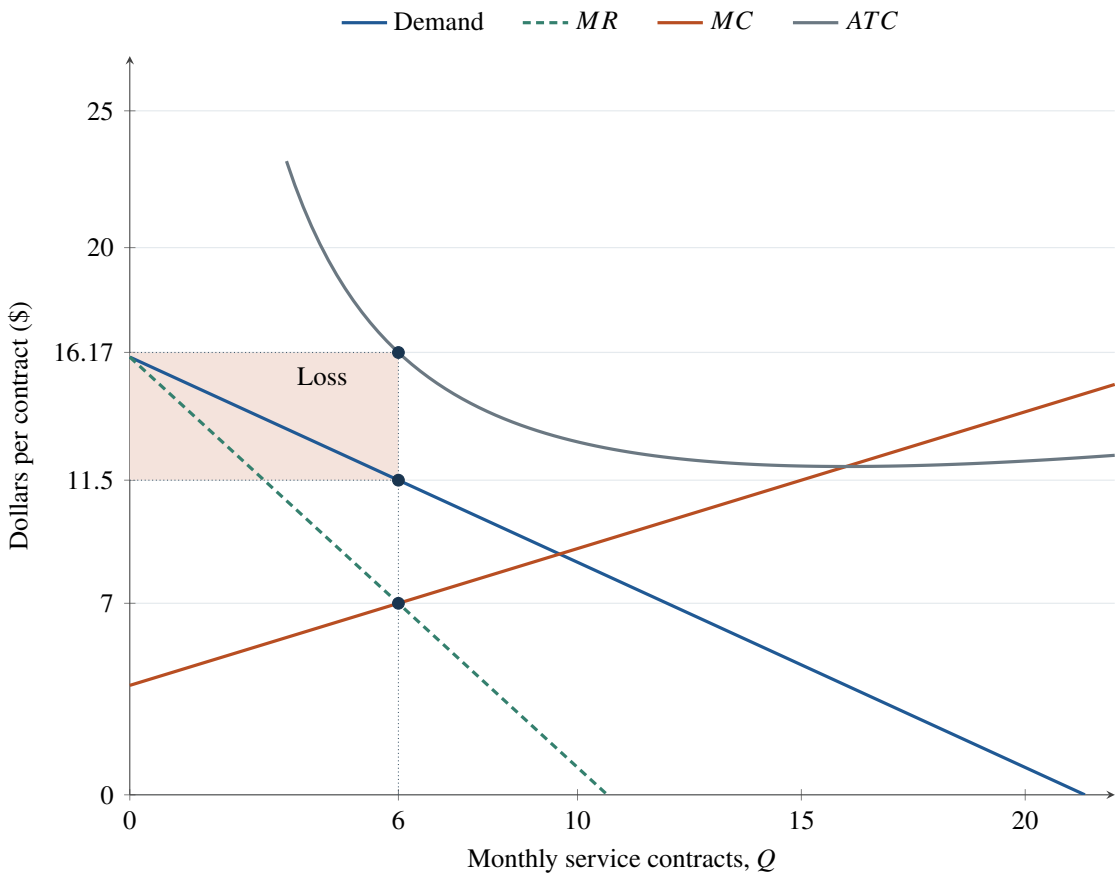


Figure 2. Short-run economic loss. The firm makes 6 contracts at \$11.50 each, with $ATC \approx \$16.17$. The shaded area is a \$28 loss; continuing is better than paying the \$64 fixed cost with no sales.

Read the graph. The best output is still set by $MR = MC$, not by the point at which demand crosses ATC. At $Q = 6$, demand lies below ATC, so the shaded rectangle is a loss. That fact alone does not establish that immediate shutdown is best.

So what? Product differentiation permits a markup, but it does not guarantee profits. A firm can be charging more than marginal cost and still lose money because it must also recover fixed and other costs.

3. Entry and exit

Economic profit attracts competitors offering additional varieties. Customers now have more alternatives, so the *demand facing each existing firm* tends to shift left. If firms are losing money, some exit; the remaining varieties attract additional customers and their demand tends to shift right. This adjustment continues until the typical surviving firm earns zero economic profit.

KEY RESULT How entry and exit affect the individual firm

Profit → entry → each firm's demand shifts left → profit falls.

Loss → exit → remaining firms' demand shifts right → losses shrink.

Unlike the perfect-competition chapter, this is a change in the demand facing *each differentiated firm*, not simply a right or left shift of an industry's identical-product supply curve.

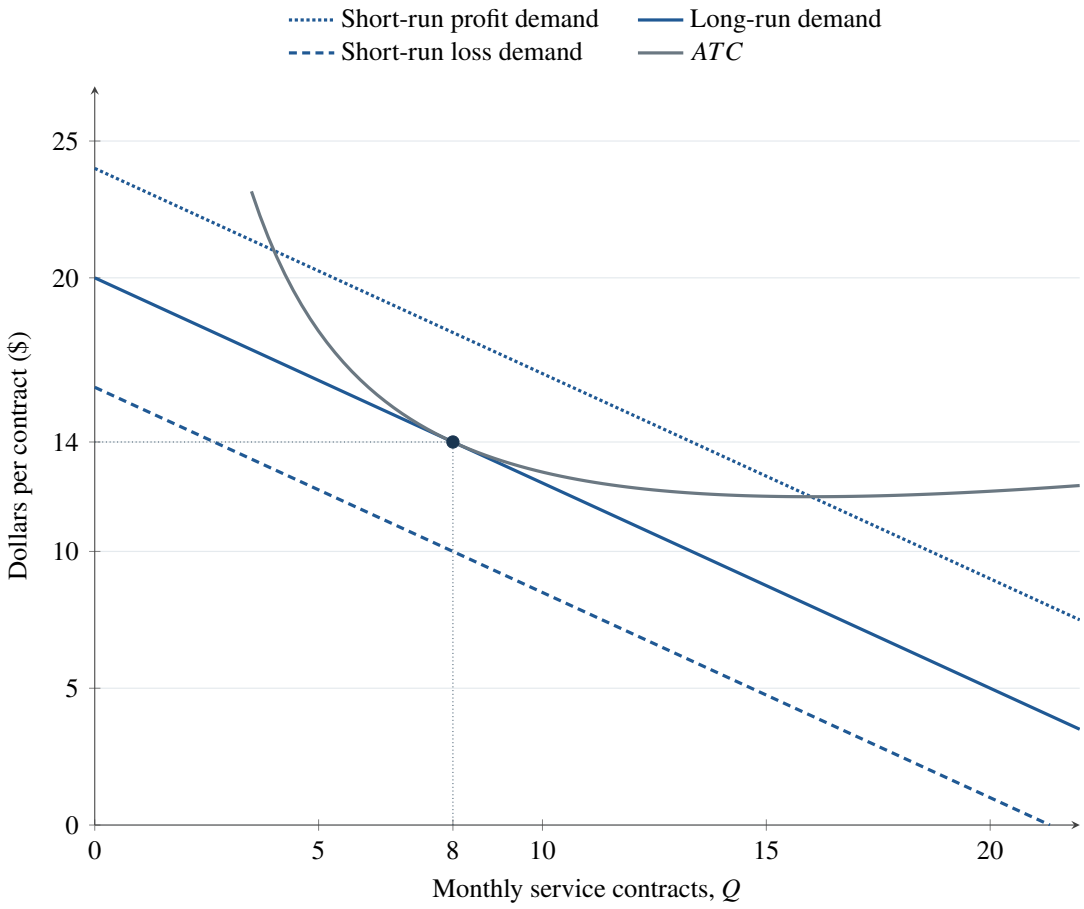


Figure 3. Entry and exit change firm-level demand. Profits induce entry and a shift from the upper demand line toward the central one; losses induce exit and a shift from the lower line toward it. The central demand curve touches ATC at (8, \$14).

Read the graph. The dotted upper line represents the profitable short-run case. As competing varieties enter, each firm's demand moves toward the solid line. The dashed lower line represents the loss-making case; exit shifts demand toward the same solid line. The exact number of firms is not determined from this single-firm graph.

The long-run outcome

With our common cost function, the long-run demand curve is $P = 20 - 0.75Q$ and $MR = 20 - 1.5Q$. At $Q = 8$, $MR = MC = \$8$, while demand gives $P = \$14$ and $ATC = 4 + 64/8 + 0.25(8) = \$14$. Thus economic profit is zero.

Notice the economic difference from perfect competition. The firm still sells at \$14 although its marginal cost is only \$8. Yet that \$6 markup is *not* economic profit per unit: average total cost is also \$14.

DEFINITION Long-run monopolistic-competition equilibrium

With free entry and exit, the typical firm earns **zero economic profit**, so $\boxed{P = ATC}$ at its chosen output. The firm also chooses $\boxed{MR = MC}$, and its downward-sloping demand curve is **tangent to** ATC on the falling portion of ATC. Thus $P > MC$ even though economic profit is zero.

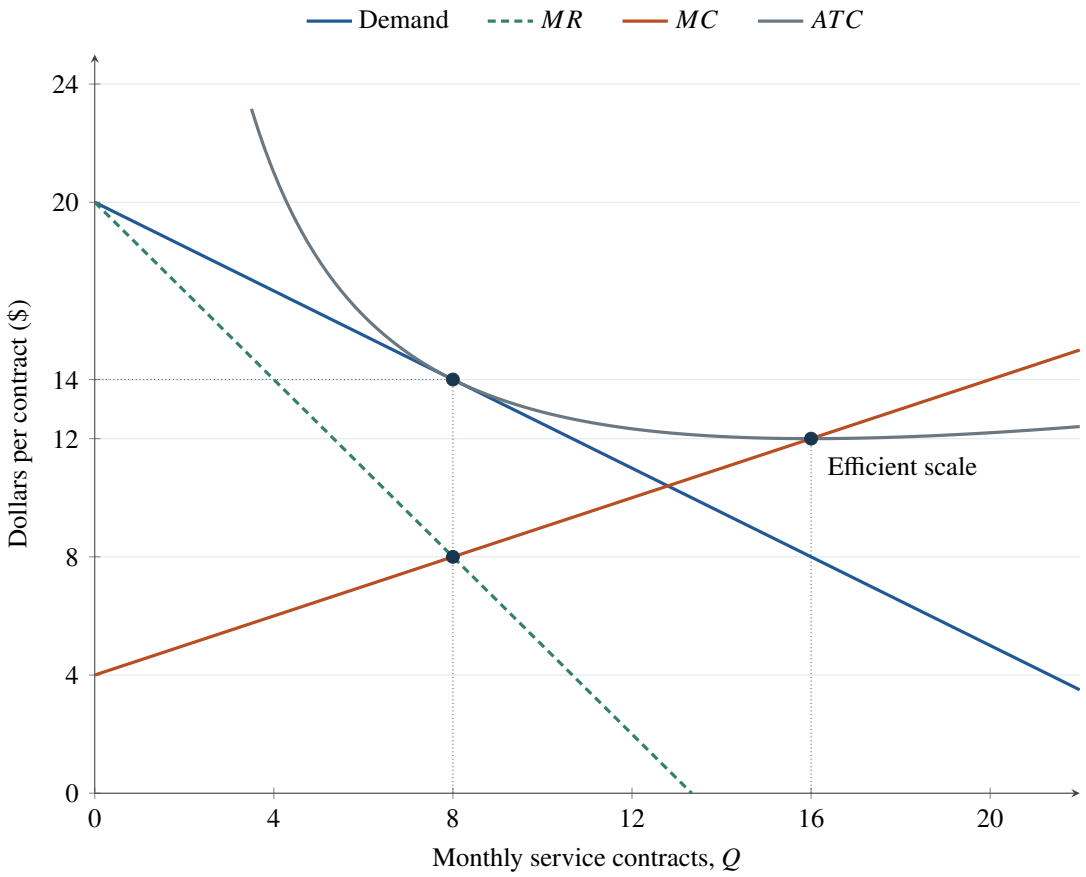


Figure 4. Long-run equilibrium of the differentiated firm. Demand just touches ATC at $Q = 8$, $P = \$14$; $MR = MC = \$8$. Minimum $ATC = \$12$ occurs at 16 contracts, beyond the firm's output.

Read the graph. The blue demand curve touches but does not cross the gray ATC curve at 8 contracts. Price and average total cost are both \$14; profit is zero. The MR and MC curves intersect directly below that same quantity at \$8. The firm's output is to the left of the minimum of ATC.

So what? Zero long-run profit does not force $P = MC$. Under monopolistic competition the markup persists because customers value product differences, while entry removes the opportunity for surplus *economic profit*.

4. Comparison with perfect competition

The key differences become visible when comparing a differentiated firm with a price-taking firm having the *same* illustrative technology. Both earn zero economic profit in their standard long-run equilibrium. But one produces below the output that minimizes average cost, while the other produces at that minimum.

> **DEFINITION Excess capacity and markup**
>
> **Efficient scale** is the firm's output where ATC is lowest. **Excess capacity** here means the difference between that output and the smaller long-run output of a monopolistically competitive firm. A **markup** means $P > MC$: the selling price exceeds the marginal cost of another unit. Neither concept, by itself, proves positive economic profit.

For our cost function, the minimum of ATC is \$12 at $Q = 16$; at that point $MC = \$12$. The monopolistically competitive firm sells only 8 contracts at \$14, leaving 8 contracts of capacity *relative to efficient scale* unused. Its markup is \$14 minus \$8, or \$6 per contract. A competitive firm with these costs instead faces a horizontal price line of \$12, chooses 16 contracts, and earns zero economic profit.

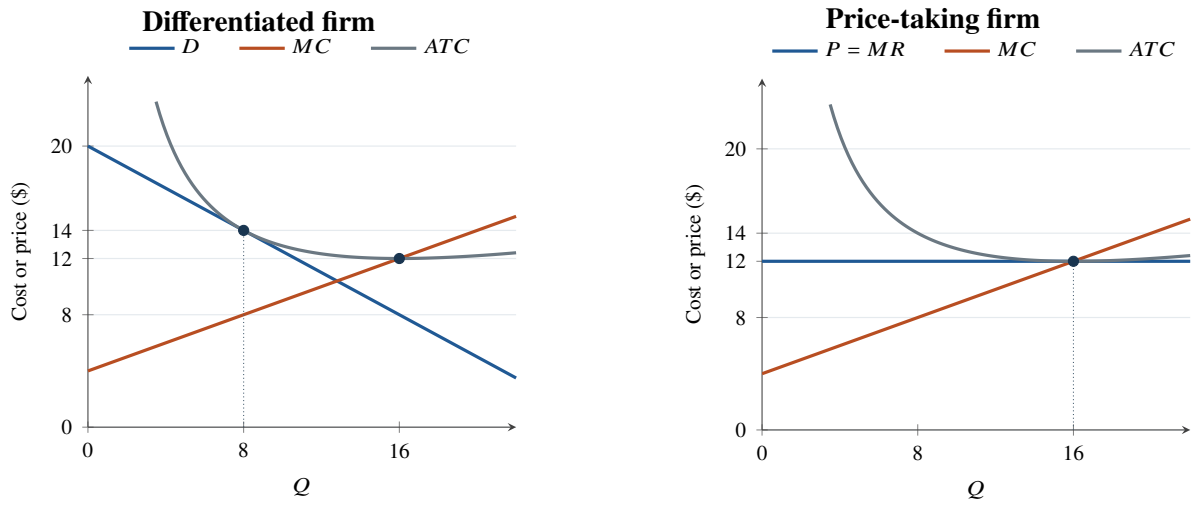


Figure 5. Long-run comparison with the same illustrative costs. Left: differentiated demand touches ATC at 8 contracts, $P = \$14 > MC = \$8$. Right: the price taker produces 16 at minimum $ATC = MC = P = \$12$.

Read the graph. Both firms earn zero economic profit because price equals average total cost. The left-hand firm is on the *falling* part of ATC and has a positive markup. The right-hand firm is at minimum ATC, where the horizontal price line crosses marginal cost. This is a comparison of firm types, not a claim that both face the same demand curve.

Long-run feature	Perfect competition	Monopolistic competition
Products	Identical	Differentiated
Individual firm demand	Horizontal	Downward sloping
Marginal revenue	$MR = P$	$MR < P$ at positive output
Output choice	$MR = MC$	$MR = MC$
Price and marginal cost	$P = MC$	$P > MC$
Economic profit	Zero with free entry	Zero with free entry
Efficient-scale output	Yes	No: excess capacity

5. Welfare

At the output selected by a differentiated firm, price exceeds marginal cost. Some customers value additional units more than the cost of producing them but do not buy at the prevailing price. That is the standard *markup distortion*. It resembles the deadweight loss under monopoly, although the welfare effects of introducing and removing product varieties complicate the overall judgment.

> **DEFINITION Deadweight loss and entry externalities**
>
> **Deadweight loss** is surplus from mutually beneficial trades that does not occur. A **product-variety externality** is the extra benefit to buyers from having another variety available, which an entrant may not capture fully. A **business-stealing externality** is the loss of some sales and profit at existing firms when an entrant attracts their customers.

Worked example: The markup leaves some mutually beneficial sales unrealized

Holding the firm's long-run demand and costs fixed, the differentiated firm chooses 8 contracts at \$14, with marginal cost \$8. Customers' willingness to pay for the next units remains above marginal cost. Demand $P = 20 - 0.75Q$ meets $MC = 4 + 0.5Q$ at $Q = 12.8$, where both equal \$10.40. The triangle between the curves from 8 to 12.8 has width $12.8 - 8 = 4.8$ and height $14 - 8 = \$6$. Its area is $\frac{1}{2}(4.8)(6) = \boxed{\$14.40}$.

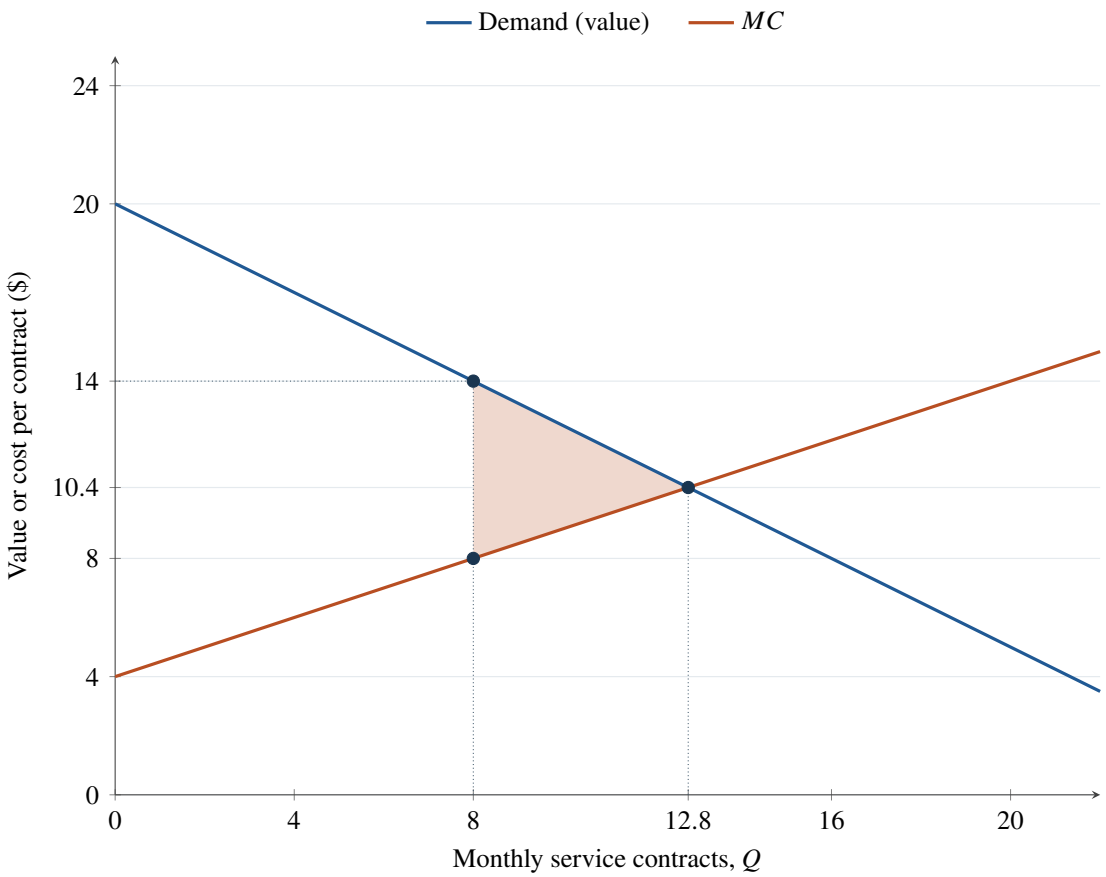


Figure 6. Illustrative markup deadweight loss, holding the differentiated variety fixed. At $Q = 8$, demand is \$14 and $MC = \$8$. Their intersection is at $Q = 12.8$, \$10.40. The closed shaded triangle represents \$14.40 of unrealized surplus.

Read the graph. Between 8 and 12.8 contracts, the blue demand curve lies above the orange marginal-cost curve. Those transactions would generate extra joint gains. The shaded area measures the unrealized gains *for this fixed variety and demand*; it does not by itself measure the total welfare effect of changing the number of competing firms.

> **KEY RESULT Free entry need not produce the best number of varieties**
>
> **Too little entry is possible** because a new variety benefits customers beyond the entrant's own profit. **Too much entry is also possible** because entrants attract purchases that would otherwise have supported established firms; they do not necessarily count that lost business when making entry decisions. The two forces work in opposite directions, so free entry alone does not tell us whether there are too many or too few firms.

So what? Zero economic profit establishes that entry incentives have disappeared, not that total welfare is maximized. For differentiated products, price, output, variety, and the costs of maintaining additional firms all matter.

6. Advertising and brand names

Differentiated firms compete for customers through product features, prices, reputation, and advertising. Because price often exceeds marginal cost, attracting one more customer can be profitable. But whether advertising benefits society is a different question from whether it benefits an individual firm.

> **DEFINITION Advertising and brand names**
>
> **Advertising** is paid communication intended to influence or inform buyers about a product. A **brand name** identifies a seller's product and can carry a reputation across repeated purchases. Both may communicate useful information, but they can also increase perceived differences between otherwise similar goods.

The critique and the defense of advertising

Critique. Some advertising mainly changes perceptions, encourages loyalty without improving quality, or makes close alternatives seem less substitutable. This can increase firms' pricing power and use resources without a comparable social benefit.

Defense. Advertising can tell buyers that a product exists, what it costs, where it is available, or how it differs from alternatives. Better information makes it easier to compare suppliers and can intensify competition. Neither conclusion follows merely from observing that a business advertises heavily.

Can costly advertising signal quality?

An advertisement need not describe every feature to convey information. Under some conditions, a firm confident of *repeat purchases* can rationally pay for an expensive introductory campaign that a low-quality seller would not recover. Buyers may then interpret the firm's willingness to make that investment as a signal. This logic requires different expected returns from repeat sales; spending money alone is not proof of quality.

Worked example: An advertising campaign and repeat service contracts

Two hypothetical scheduling-software suppliers consider a \$12,000 campaign. Each gains 400 trial customers, contributing \$12 per customer-month *after variable service costs*. Both earn $400(12) = \$4,800$ initially. Over the next 8 months, a reliable product retains 60% of those customers; an unreliable product retains only 10%. Suppose retention stays at the stated fraction over those 8 months.

	Reliable product	Unreliable product
Initial contribution	\$4,800	\$4,800
Repeat customers	240	40
Additional contribution	$240(8)(12) = \$23,040$	$40(8)(12) = \$3,840$
Less advertising cost	$-\$12,000$	$-\$12,000$
Net contribution from campaign	**\$15,840**	$-\$3,360$

The reliable supplier has an incentive to advertise because its campaign generates more contribution than it costs. The unreliable supplier would lose money by copying that campaign under the stated assumptions. **So what?**

Costly promotion can sometimes convey private information about quality, but it does not always do so; firms might advertise for other reasons and buyers can be mistaken.

Brand names: the critique and the defense

Critique. Buyers may pay more for a familiar name even when another product performs just as well. Heavy brand promotion can reduce attention to price and strengthen market power.

Defense. A recognized name can reduce the time spent investigating unfamiliar sellers. A firm with valuable reputation has an incentive to maintain quality because poor performance could damage future sales. The economic question is whether the information and incentives outweigh any extra price and marketing costs.

> **KEY RESULT Advertising has no unconditional welfare verdict**
>
> Advertising and branding can improve matching, information, and quality incentives. They can also waste resources or soften price competition. Judge their effects by what buyers learn, how products and prices change, and whether the resulting benefits exceed the real resources spent.

Quick review: identify the rule

Question	Economic explanation or test
Market conditions?	Many sellers; differentiated products; free entry and exit.
Firm's output and price?	Choose $MR = MC$; read price from demand.
Profit or loss?	$(P - ATC)Q$ at chosen output.
Effect of entry?	Existing firms' demand shifts left, reducing profit.
Long-run equilibrium?	Demand tangent to ATC; $P = ATC$; zero economic profit.
Excess capacity?	Firm output is below minimum-ATC output.
Why a markup?	Because $MR < P$, the firm sets $P > MC$ at $MR = MC$.
Is free entry efficient?	Variety benefits and business stealing oppose one another.
Role of advertising?	It can inform, signal quality, or waste resources.

Three common mistakes, corrected

Mistake 1: "Zero long-run profit implies price equals marginal cost." False. In our long-run example, $P = ATC = \$14$, but $MC = MR = \$8$. The \$6 markup pays toward the firm's average costs; it is not economic profit.

Mistake 2: "A firm facing a downward-sloping demand curve must be a monopoly." False. Differentiated firms have downward-sloping individual demand even when numerous rivals offer competing varieties and entry is free.

Mistake 3: "Free entry guarantees the efficient number of firms." False. Entry adds consumer value through product variety but can also displace business at existing firms. Without measuring both effects, the efficient number of varieties is unknown.

> **KEY RESULT One final connection**
>
> A monopolistically competitive firm behaves like a monopolist when choosing short-run output: $MR = MC$, with $P > MR$. It behaves like a competitive industry in one crucial long-run respect: free entry and exit eliminate economic profit. Together, these imply $P = ATC > MC$, excess capacity, and a differentiated-product market whose entry and advertising effects require separate economic judgment.